

This alarm dataset is based simulation dataset made openly available in [1]. The original dataset from [1] compiles a large array of abnormal situations simulated in the PCTTRAN nuclear power plant (NPP) accident simulation software. PCTTRAN is a state-of-the-art simulator for abnormal situations and accidents in NPPs, e.g. used as a benchmark simulation model for the “International Atomic Energy Agency’s” (IAEA) yearly “Advanced NPP Simulation Workshop” [2]. The original dataset [1] is intended for study of accidents in NPPs by data scientists, but did not provide any alarm data. This dataset contains a comprehensive alarm dataset created from [1] intended to be used as a benchmark, in order to facilitate comparable research in the field of alarm management.

The dataset contains alarm time series for each of the 1211 samples from [1] divided into 12 different fault classes. To facilitate research in to the effects of different alarm system configurations, or e.g. and make algorithms more robust to badly configured alarm systems, a total of 101 alarm time-series is provided for each of the original 1211 samples, resulting in 122.412 series in total. These 100 time-series comprise different alarm system configurations, generated through varying tradeoffs between missed alarm rate (MAR) and false alarm rate (FAR). We defined the MAR for each variable as the percentage of datapoints in a disturbed state, where the alarm of that variable is not triggered and the FAR as the percentage, where an alarm is triggered in absence of a disturbance. This tradeoff was calculated as:

$$C_i(\alpha, \tau) = \alpha \times MAR_i(\tau) + (1 - \alpha) \times FAR_i(\tau),$$

where τ is the process variable, α is the tradeoff parameter and C_i is the score. For each alpha and variable, the high- and low-alarm limits with the greatest score were selected. In cases where multiple alphas achieved the same best score the lowest high-alarm limit or respectively the highest low alarm-limit with the same score were selected to achieve maximum coverage of disturbed states.

Alarm-Limit files:

These CSVs are not part of the alarm data, but contain the high- and low-alarm limits, the proportion of all datapoints from the steady and disturbed state, that trip these limits (for high, low and total). There is one file for each alpha. The order of rows in the CSV is as follows: 1. Variable name, 2. High-alarm limit, 3. High-alarm limit score 4. Steady state proportion active with high-alarm, 5. Disturbed state proportion active with high-alarm, 6. Low-alarm limit, 7. Low-alarm limit score, 8. Steady state proportion active with low-alarm, 8. Disturbed state proportion active with low-alarm, 9. Steady state proportion active with any alarm, 10. Disturbed state proportion active with any alarm.

Alarm-Data files:

For each value of alpha there exists a folder (e.g. for an alpha value of 0.01 the folder is called \1\...), that contains the corresponding alarm data for all disturbance situations (e.g. \1\FLB\... etc.) and the steady state (“normal”).

Notes on changes compared to [1]:

Compared to [1] five of the 17 original fault classes, where dropped, as they only contained a single sample, making them unsuitable for classification. However, with the provided alarm limits the according time series can be generated for these classes, if necessary. The variables *WPCS*, *WPMU* and *WPFW* where only contained in the SLBIC disturbance. These have been removed from the dataset in order to prevent problems with imbalanced variables over the different disturbance classes.

REFERENCES

- [1] B. Qi, X. Xiao, J. Liang, L.C. Po, L. Zhang, and J. Tong, “An open time-series simulated dataset covering various accidents for nuclear power plants,” *Scientific Data*, vol. 9, article 766, Dec. 2022, doi: 10.1038/s41597-022-01879-1.
- [2] L.C. Po, “Conceptual design of an accident prevention system for light water reactors using artificial neural network and high-speed simulator,” *Nuclear Technology*, vol. 206, no. 3, pp. 505–513, Aug. 2019, doi: 10.1080/00295450.2019.1641877.